Linear System

$y = Ax$ $M<N$ 欠定 无解或无穷解 $y \in range(A)$

$rank(A) = rank(A|y)$

$rank(A) = dim(row(A)) \leq M$

AA^H 可逆，若 A 满秩 $\hat{x}_{LS} = A^H(AA^H)^{-1}y$

A^HA 必不可逆

无解时，$A^H(AA^H)^{-1}y$ 是最接近的

无穷解时，$\underbrace{A^H(AA^H)^{-1}}_{A^\dagger}y$ 是众多解之一

$y = Ax$ $M>N$ 过定 无解或唯一解 $y \in range(A)$

$rank(A) = rank(A|y)$

$rank(A) = dim(col(A)) \leq N$

A^HA 可逆，若 A 满秩 $\hat{x}_{LS} = (A^HA)^{-1}A^Hy$

AA^H 必不可逆

无解时，$(A^HA)^{-1}A^Hy$ 是最接近的

唯一解时，$\underbrace{(A^HA)^{-1}A^H}_{A^\dagger}y$ 即唯一解

LS

$\min \ \|y - Ax\|_2^2$

$\nabla f = 2A^H(y - Ax) = 0$

$\hat{x}_{LS} = (A^HA)^{-1}A^Hy$ 为什么求导得不出 $(A^HA)^{-1}A^Hy$

Ridge

$\min \ \|y - Ax\|_2^2 + \lambda\|x\|_2^2$

$\min \ \|\underbrace{\begin{pmatrix} y \\ 0 \end{pmatrix}}_{\tilde{y}} - \underbrace{\begin{pmatrix} A \\ \sqrt{\lambda} I \end{pmatrix}}_{\tilde{A}} x\|_2^2 = \|\tilde{y} - \tilde{A}x\|_2^2$ $\hat{x}_{ridge} = (\tilde{A}^H\tilde{A})^{-1}\tilde{A}^H\tilde{y}$

$\hat{x}_{ridge} = (A^HA + \lambda I_N)^{-1}A^Hy = A^H(AA^H + \lambda I_M)^{-1}y$

Pf: inverse lemma $(A^HA + \lambda I_N)^{-1}A^H$

$= [\frac{1}{\lambda}I_N - \frac{1}{\lambda}A^H(A \cdot \frac{1}{\lambda}I_N \cdot A^H + I_M)^{-1}A\frac{1}{\lambda}]A^H = \frac{1}{\lambda}A^H - \frac{1}{\lambda}A^H(AA^H + \lambda I_M)^{-1}AA^H$

$= \frac{1}{\lambda}A^H(AA^H + \lambda I_M)^{-1}[(AA^H + \lambda I_M) - AA^H] = A^H(AA^H + \lambda I_M)^{-1}$

Psedu inverse

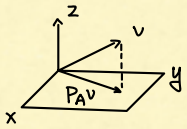
$A^\dagger = (A^HA)^{-1}A$ $M>N$ 过定，$rank(A) = dim(col(A)) = N$

$A^\dagger = A^H(AA^H)^{-1}$ $M<N$ 欠定，$rank(A) = dim(row(A)) = M$

Projection

$\mathrm{Proj}_A = A(A^H A)^{-1} A^H$ 向A列空间的投影

$\mathrm{Proj}_A^{\perp} = I - A(A^H A)^{-1} A^H$



e.g. $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}$ $\mathrm{Proj}_A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$